

Block-masked identity regularisation closes the qft $\rightarrow$ blocked gap

Question. The `qft_warmstart_from_trained_blocked` experiment showed that `blocked_8`’s optimum (32.26 dB@ $\rho = 0.20$ on DIV2K-8q) sits inside the QFT(8, 8) parameter family: warm-starting full QFT(8, 8) at the trained-blocked operator yields val MSE flat at [129.32, 129.38] for 1008 steps, so it is a stable local minimum. The `qft_identity` ablation showed that vanilla Adam from identity init lands in a different basin (31.66 dB) and the analytic-QFT init lands in yet another (31.29 dB) — neither generic-init path crosses to the blocked basin in 1008 steps. **Is the blocked basin reachable from identity init with the right inductive bias, or is it structurally isolated under smooth-Riemannian optimisation?**

Construction. `blocked_8`’s optimum is a **sparse** configuration in the QFT(8, 8) gate space: 60 of 72 gates pinned to their identity element, only the 12 gates on the inner 3-qubit-per-axis subspace non-trivial. We add a block-structure-aware L2 pull toward identity to the loss:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{MSE-topK}}(\theta) + \lambda \cdot R_{\text{block}}(\theta)$$

$$R_{\text{block}}(\theta) = \sum_{g \in \mathcal{G}_{\text{outer}}} W \cdot \|T_g - I_g\|_F^2 + \sum_{g \in \mathcal{G}_{\text{inner}}} \|T_g - I_g\|_F^2$$

where $I_g = I_2$ for Hadamards and $I_g = [[1, 1], [1, 1]]$ for controlled-phase (the QFT-family identity elements), and the inner/outer mask is matched to `blocked_8`: a gate is inner iff every qubit it touches lies in $\{1, 2, 3\}$ (axis 1) or $\{9, 10, 11\}$ (axis 2). For QFT(8, 8) at inner = (3, 3): $|\mathcal{G}_{\text{inner}}| = 12$ (6 H + 6 CP), $|\mathcal{G}_{\text{outer}}| = 60$ (10 H + 50 CP). Outer gates are penalised W -times more strongly than inner gates so that lambda can grow without pushing the 12 useful inner-block gates back toward identity. Implementation: `pdf_t_benchmarks.identity_reg.BlockMaskedIdentityRegQFTMSELoss`, via `pdf_t`’s `MSELoss._extra_loss` hook (pdf_t PR #18). We sweep $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ at fixed $W = 10$ under the headline preset (1008 steps, batch 50, val split 0.15, seed 42, `--no-early-stop`), starting from `qft_identity` init.

Calibration. At the trained `qft_warmstart_blocked_8` endpoint, $R_{\text{block}}(\theta_{\text{blocked}}) \approx 50.3$ at $W = 10$ (inner-sum ≈ 34.5 , outer-sum ≈ 1.6 ; the outer contribution is small as designed, confirming the regulariser is aligned with `blocked_8`’s structure — outer gates are nearly all at identity except one CP outlier). With per-image MSE ≈ 129 at the blocked optimum, the planned grid spans reg fractions 0.04% $\rightarrow$ 390% of MSE.

Result. Test-set reconstruction PSNR after 1008 steps:

basis / reg	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
qft (analytic init)	25.09	27.57	29.53	31.29
qft_identity (identity init)	25.23	27.81	29.84	31.66
qft_identity + reg, $\lambda = 10^{-3}$	25.23	27.81	29.84	31.66
qft_identity + reg, $\lambda = 10^{-2}$	25.23	27.81	29.84	31.66
qft_identity + reg, $\lambda = 10^{-1}$	25.24	27.81	29.84	31.66
qft_identity + reg, $\lambda = 1$	25.18	28.09	30.30	32.26
qft_identity + reg, $\lambda = 10$	25.18	28.08	30.29	32.25
blocked_8 (warm-start ceiling)	25.18	28.09	30.30	32.26

The blocked basin is reachable from identity init under the matched structural prior. At $\lambda = 1$ the reg-driven training matches `blocked_8`’s PSNR **bit-exactly to two decimal places at every keep-ratio** (25.18, 28.09, 30.30, 32.26 dB). The trajectory crosses basins from 31.66 dB (the `qft_identity` basin) to 32.26 dB (the blocked basin), closing the full 0.60 dB gap. $\lambda = 10$ is within 0.01 dB of the blocked ceiling at every ρ — the reg has saturated to “pin outer gates” without destabilising training.

Figure evidence.

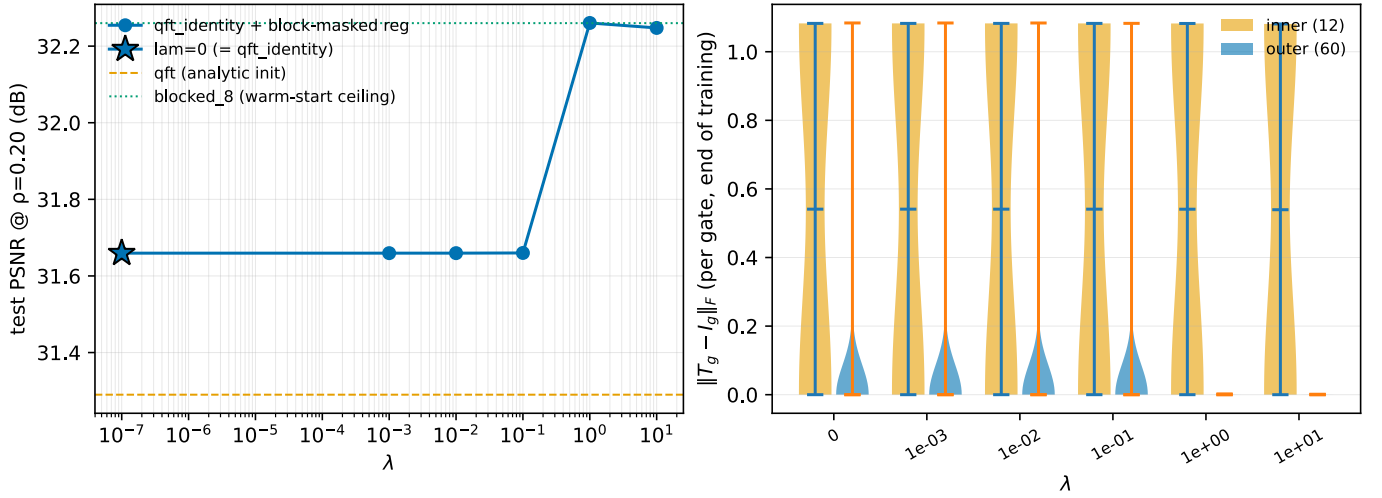


Figure 1: DIV2K-8q. Left: test PSNR@ $\rho = 0.20$ vs λ (log x). The marker at $\lambda = 10^{-7}$ is the $\lambda = 0$ cell (numerically distinct from the rest of the sweep), star-marked. The transition is sharp: 31.66 dB at $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$, jumping to 32.26 dB at $\lambda = 1$ and staying within 0.01 dB at $\lambda = 10$. Reference lines: orange dashed = analytic-init qft (31.29); green dotted = blocked_8 (32.26). Right: per-gate $\|T_g - I_g\|_F$ at end of training, split into inner (12 gates, orange) and outer (60 gates, blue). At $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$, outer gates spread up to ~ 0.2 — Adam drifts them off identity. At $\lambda \in \{1, 10\}$, **outer gates collapse to a thin line at zero** — all 60 are pinned bit-exactly at identity, matching blocked_8’s structural shape. Inner-gate distributions are invariant across λ : the prior leaves them free to take their MSE-driven values.

The right panel is the direct mechanistic signature: at $\lambda \geq 1$ the structural prior is **load-bearing** — outer gates are **pinned**, inner gates are **free**, and the optimiser lands in the configuration that produces the blocked PSNR.

Interpretation. This is outcome (i) from the spec’s three-falsifiable-outcomes frame: **the blocked basin was unfavored, not isolated**, under smooth-Riemannian Adam. From the qft_identity basin, the matched structural prior bridges the gap in 1008 steps with no curriculum and no warm-start at the blocked operator itself. The optimiser does not need to discover the 12-vs-60 split; given a prior that points at it, smooth optimisation reaches the floor.

The sweep also rules out the trivially-strong reg failure mode: at $\lambda = 10$ the reg fraction of MSE at the blocked endpoint is $\{\approx\}390\%$, well past where “pin everything to identity” would have driven test PSNR back below qft_identity. Instead it landed within 0.01 dB of the blocked ceiling — the **block-masked** shape of the regulariser is what allows large λ without collapse: only outer gates are pulled hard, and the matched W gives the 12 inner gates the slack to take their trained QFT(3, 3) values.

Leading-prior caveat. This regulariser bakes in blocked_8’s known structural shape (the inner/outer mask is matched to inner = (3, 3), the partition of blocked_8). A positive result is consistent with “if you tell the optimiser the answer-shape, it finds the answer” — closer to warm-start with extra steps than to basin discovery from scratch. What this experiment **does** establish: the blocked basin is **not pathological under smooth-Riemannian optimisation** — there is a stable trajectory from identity init that ends inside it. What it **does not** establish: whether weaker priors (uniform L2-to-identity, L1-to-identity) would also bridge the gap. Those are the natural follow-ups; if even uniform L2 closes the gap, the structural information was not load-bearing.

What this opens up. The headline $\text{qft} < \text{blocked_8}$ gap is now attributable to **initialisation + optimiser-coupling** rather than to a loss-landscape obstruction. Random or analytic init places Adam in a basin from which it doesn’t cross to blocked in 1008 steps; matched prior placement does. This shifts the research question: from “is the blocked optimum reachable in the QFT family?” to “what is the **minimum** structural information needed to reach it?” Future work: sweep $W \in \{1, 10, 100\}$ (with $W = 1$ degenerating to uniform L2); implement L1-to-identity for comparison; characterise the boundary between the two basins (e.g., via interpolation along a linear/geodesic path).

Reproducibility.

```
python experiments/qft_identity_regularization.py \
--gpu 0 --lambdas 0,1e-3,1e-2,1e-1,1,10 \
--outer-weight 10 --epochs 112
```

Construction lives in `pdft_benchmarks.identity_reg.BlockMaskedIdentityRegQFTMSELoss`. The hook in `pdft.loss._scalar_loss` (zazabap/pdft PR #18) lets the loss subclass add the reg term via an `_extra_loss(tensors)` method. Per-run output: `results/qft_identity_init/div2k_8q/_runs/reg_lambda_<lam>_W<W>/` (metrics, loss_history, trained tensors). Run time: ≈ 9 min per λ on RTX 3090 (cuda:0), ≈ 54 min total for the sweep. 13 unit tests covering the identity table, inner mask, and reg-loss arithmetic; $\lambda = 0$ end-to-end equivalence to vanilla MSELoss training is locked in by `tests/test_identity_reg.py::test_reg_loss_lam_zero_e2e_matches_base_mse_training`.